

VIBRATION SENSING FOR PREDICTIVE MAINTENANCE

The introduction of fully-integrated and reliable vibration sensors can enable autonomous and configurable operation. With these, predictive maintenance program developers can significantly improve the quality and integrity of the data collection process without the limitations and compromises posed by past vibration analysis approaches



3-axis vibration analyser with embedded FFT processing

■ BOB SCANNELL

Precision industrial processes are increasingly reliant on efficient and consistent operation of motors and associated machinery. Imbalances, defects, loose fittings and other anomalies in the machinery typically translate into vibration, resulting in loss of precision as well as safety concerns. If left unaddressed, besides the performance and safety issues, loss of productivity becomes inevitable if

equipment needs to be taken-off the line for repair.

Condition-based predictive maintenance is a known and proven approach for avoiding productivity loss. But this approach has its complexities. Existing methods have limitations, particularly when it comes to analysing the vibration data and isolating error sources.

Existing data collection approaches include simple piezo-based sensors mounted to the machinery and handheld data collection tools. These

methods have a number of drawbacks, particularly when compared with the ideal solution of a complete detection and analysis system, which can be embedded in the machinery and act autonomously.

The introduction of fully-integrated and reliable vibration sensors have the ability for autonomous and configurable operation. With these, predictive maintenance program developers can significantly improve the quality and integrity of the data collection process, without the limitations and compromises posed by past vibration analysis approaches.

Repeatability of measurements

Handheld vibration probes offer some implementation advantages. These do not require any modification to the end-equipment, and given their large (brick) size, these are relatively highly integrated, allowing sufficient processing and storage. However, a major limitation is the repeatability of the measurements. Slight differences in the probe location or angle produces inconsistent vibration profiles making time comparisons inaccurate. Thus, the maintenance technician is left with the question of whether any observed vibration shift is due to an actual change within the machinery, or just a change in the measurement technique. Ideally the sensor would be both compact and integrated sufficiently to allow direct